

Statistics Project

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January 2026

1 Introduction

Frailty is theoretically defined as a clinically recognizable state of increased vulnerability resulting from aging-associated decline in reserve and function across multiple physiologic systems such that the ability to cope with everyday or acute stressors is comprised [1]. It has been considered a predictor of disability, hospitalization, falls, loss of mobility, cardiovascular disease and death [2]. Hemodialysis patients are exposed to recurrent metabolic stress, dietary restrictions, and treatment-related inflammation, which increase vulnerability to frailty, particularly in older adults. Frailty can be quantified using the Multidimensional Prognostic Index (MPI), a validated prognostic score designed to summarise global patient vulnerability in elderly populations [3].

Malnutrition plays a key role in the progression of frailty in this population. Patients on maintenance hemodialysis are at high risk of protein–energy wasting, characterised by reduced dietary intake, muscle loss, and altered metabolism. For the assessment of nutritional status, we adopt the Global Leadership Initiative on Malnutrition (GLIM), a consensus-based framework that allows malnutrition to be identified using clinically relevant phenotypic indicators (e.g. weight loss or muscle depletion) and etiologic factors (e.g. reduced food intake or chronic disease burden) [4].

The objective of this research project is to examine how nutritional and clinical characteristics, including GLIM-defined malnutrition status, muscle and fat mass, serum albumin concentration, diabetes, dialysis adequacy, age, and sex, contribute to variation in frailty as measured by the MPI score. The research question guiding this study is: How do nutritional and clinical factors influence frailty in elderly hemodialysis patients? We employ multiple linear regression to evaluate the relationship between frailty and several nutritional and clinical factors simultaneously, and perform a statistical comparison of frailty levels between malnourished and non-malnourished patients.

2 Dataset & Variables

The analysis is based on an anonymised clinical dataset provided by a local hospital, comprising elderly patients aged 65 years and older undergoing maintenance hemodialysis. The dataset includes demographic, clinical, and nutritional variables collected during routine clinical assessment. Each observation corresponds to a single patient. Due to confidentiality constraints, the dataset is not publicly available but is provided as part of the project submission for academic evaluation [5].

Frailty is the dependent variable of the analysis and is operationalised using the Multidimensional Prognostic Index (MPI), which is recorded as a continuous variable ranging from 0 to 1, with higher values indicating greater frailty. Nutritional status is captured through the Global Leadership Initiative on Malnutrition (GLIM) classification,

included in the analysis as a binary indicator of malnutrition. Additional nutritional variables include lean tissue index (LTI), fat mass percentage, and serum albumin concentration, all treated as continuous covariates. The set of clinical and demographic covariates further includes age, sex, diabetes status, and dialysis adequacy, measured using the KT/V index.

Prior to the statistical analysis, we selected the variables relevant to the research question and excluded observations with missing values for the dependent variable or key covariates. To obtain an initial understanding of the data structure and potential relationships, we conducted a preliminary exploratory analysis by visualising the distribution of frailty and its pairwise associations with nutritional and clinical predictors. This exploratory step informed subsequent modelling choices and provided insight into general patterns present in the data.

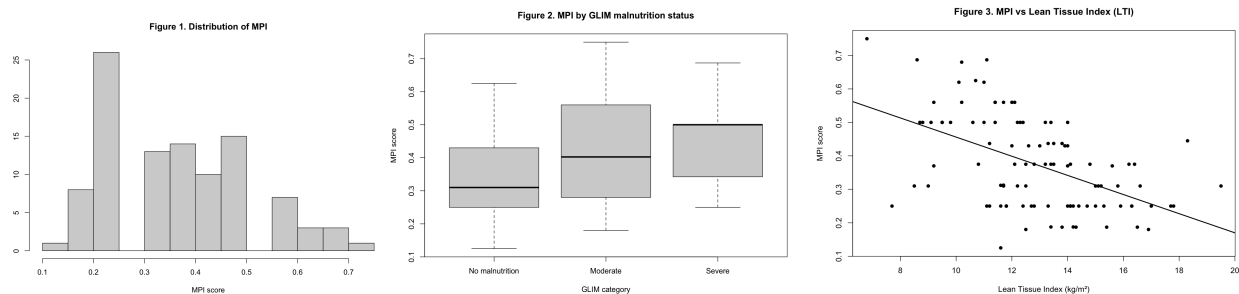
3 Data Cleaning and Exploratory Analysis

This section examines the dataset prior to regression analysis, focusing on the main characteristics of the study population and the distribution of frailty.

Table 1. Characteristics of the study population

Variable	Value
Number of patients	101
MPI score	0.37 $\pm$ 0.14
Age (years)	77.58 $\pm$ 6.47
Lean Tissue Index (kg/m ²)	12.86 $\pm$ 2.49
Fat mass (%)	33.49 $\pm$ 9.03
Albumin (g/dL)	3.86 $\pm$ 0.32
KT/V	1.40 $\pm$ 0.25
CIRS score	13.43 $\pm$ 3.17
GLIM classification – No malnutrition	61 (60.4%)
GLIM classification – Moderate malnutrition	20 (19.8%)
GLIM classification – Severe malnutrition	20 (19.8%)
Sex – Male	81 (80.2%)
Sex – Female	20 (19.8%)

Descriptive characteristics of the study population are reported in Table 1.



Prior to the analysis, observations with missing values for the dependent variable or key covariates were excluded, resulting in a final sample of 101 patients (down from 123). Descriptive statistics (Table 1) indicate substantial heterogeneity in the study population with respect to frailty, nutritional status, and clinical characteristics. The distribution of the MPI score (Figure 1) shows a wide spread across the frailty scale, with patients distributed from lower to higher vulnerability levels rather than clustering around a single value.

The boxplot comparing MPI across GLIM malnutrition categories (Figure 2) suggests a clear gradient in frailty, with higher MPI values observed among patients classified as moderately or severely malnourished compared to those without malnutrition. Although overlap between groups is present, the central tendency of frailty appears progressively higher with increasing malnutrition severity.

Furthermore, the scatterplot of MPI against lean tissue index (Figure 3) indicates an inverse pattern, whereby lower muscle mass is generally associated with higher frailty scores, while patients with higher LTI tend to exhibit lower MPI values.

4 Regression Analysis

To examine the joint relationship between frailty and multiple nutritional and clinical factors, we estimate a multiple linear regression model with the MPI score as the dependent variable. Given the continuous nature of the outcome and the aim of assessing the independent contribution of each predictor while controlling for the others, multiple linear regression provides an appropriate and interpretable framework for the analysis. The set of explanatory variables includes GLIM malnutrition status, lean tissue index, fat mass percentage, serum albumin concentration, dialysis adequacy (KT/V), age, sex, and comorbidity burden measured by the CIRS score.

Results from the multiple linear regression analysis are reported in Table 2. Overall, the model explains a large proportion of the variability in frailty, with an adjusted R^2 of 0.41, indicating good explanatory power for a cross-sectional clinical dataset. Lean tissue index is strongly and negatively associated with frailty, suggesting that lower muscle mass is linked to higher MPI scores. Fat mass percentage also shows a statistically significant negative association with frailty, indicating that patients with lower energy reserves tend to exhibit higher vulnerability. Similarly, serum albumin concentration is inversely associated with frailty, consistent with poorer nutritional status being related to increased frailty.

Age displays a weak positive association with frailty at the margin of statistical significance, while GLIM malnutrition status, dialysis adequacy, sex, and comorbidity burden are not significantly associated with frailty once other variables are accounted for in the model. Taken together, these results suggest that body composition and biochemical markers of nutrition play a central role in explaining frailty among elderly hemodialysis patients, even after adjusting for clinical and demographic characteristics.

Variable	Coefficient	Std. Error	p-value
Intercept	0.919	0.316	0.004
GLIM malnutrition	0.014	0.016	0.393
Lean Tissue Index (kg/m ²)	−0.032	0.008	<0.001
Fat mass (%)	−0.004	0.002	0.019
Albumin (g/dL)	−0.083	0.037	0.025
KT/V	0.001	0.048	0.982
Age (years)	0.004	0.002	0.061
Sex	−0.021	0.031	0.503
CIRS score	0.005	0.004	0.138

5 Group Comparison Analysis

To further examine differences in frailty by nutritional status, we conducted an independent two-sample t-test comparing mean MPI scores between malnourished and non-malnourished patients, as defined by the GLIM classification. Given the potential for unequal variances between groups, the Welch version of the t-test was employed. The null hypothesis tested was that mean frailty levels are equal across the two groups, against the alternative hypothesis that they differ.

The results indicate a statistically significant difference in mean frailty between malnourished and non-malnourished patients (mean MPI: 0.44 vs 0.33, respectively), with a test statistic of $t = 4.12$ and a corresponding p-value < 0.001 . The 95% confidence interval for the difference in means excludes zero, providing further evidence against the null hypothesis. This analysis reveals a strong unadjusted association between malnutrition and frailty, which complements the multivariable regression results by highlighting the role of nutritional status prior to adjustment for additional covariates.

6 Model Diagnostics

To assess the adequacy of the linear regression model, we performed standard diagnostic checks based on visual inspection of the residuals. In particular, we examined the relationship between residuals and fitted values, the normal Q–Q plot of residuals, the scale–location plot, and the residuals versus leverage plot.

The residuals versus fitted values plot did not reveal systematic patterns or pronounced curvature, suggesting that the assumption of linearity is reasonable and that residuals are centred around zero across the range of fitted values. The scale–location plot indicated a relatively stable spread of residuals, providing no strong evidence of heteroscedasticity.

The normal Q–Q plot showed that residuals are approximately normally distributed, with only minor deviations observed at the tails. Such departures are not unexpected in cross-sectional clinical data and are unlikely to materially affect inference given the sample size. Finally, inspection of the residuals versus leverage plot did not identify observations with excessive leverage or influence, as no points exceeded conventional Cook's distance thresholds.

7 Discussion & Conclusion

Using both exploratory analysis and multivariable regression, we found that body composition and biochemical markers of nutrition are closely associated with frailty levels in this population. In particular, lean tissue index emerged as the strongest predictor of frailty, with lower muscle mass consistently associated with higher MPI scores. Fat mass percentage and serum albumin concentration were also inversely related to frailty, suggesting that reduced energy reserves and poorer nutritional status are linked to increased vulnerability.

The group comparison analysis further showed that malnourished patients exhibit significantly higher frailty levels compared to non-malnourished patients when considered in isolation. However, in the multivariable regression framework, GLIM malnutrition status was no longer significantly associated with frailty after controlling for body composition and other covariates. This contrast highlights the difference between unadjusted and adjusted associations and suggests that the effect of malnutrition on frailty may be largely mediated through more specific nutritional and functional measures, such as muscle mass and albumin levels.

Several limitations of this study should be acknowledged. First, the analysis is based on cross-sectional data, which prevents any causal interpretation of the observed associations. Secondly, the sample size is moderate and derived from a single clinical center, which may limit the generalizability of the findings. In addition, the use of complete-case analysis led to the exclusion of some observations due to missing body composition measurements, potentially introducing selection bias.

Finally, although the MPI provides a comprehensive measure of frailty, it remains a composite index and may not capture all dimensions of vulnerability relevant to hemodialysis patients.

Despite these limitations, the analysis demonstrates the value of applying multivariable statistical methods to clinical data in order to disentangle the relative contribution of nutritional and clinical factors to frailty. Future research could build on this work by using longitudinal data to examine changes in frailty over time, expanding the set of functional and inflammatory markers considered, and evaluating the impact of targeted nutritional or physical interventions. Overall, this study illustrates how regression-based approaches can be used to meaningfully explore complex clinical relationships in elderly patient populations.

8 References

- [1] Xue QL. The frailty syndrome: Definition and natural history. *Clinics in Geriatric Medicine*. 2011 Feb;27(1):1–15.
- [2] Morley JE, Vellas B, Abellan van Kan G, Anker SD, Bauer JM, Bernabei R, et al. Frailty Consensus: A Call to Action. *Journal of the American Medical Directors Association* [Internet]. 2013 Jun;14(6):392–7. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4084863/>

[3] Pilotto A, Ferrucci L, Franceschi M, D'Ambrosio LP, Scarcelli C, Cascavilla L, et al. Development and Validation of a Multidimensional Prognostic Index for One-Year Mortality from Comprehensive Geriatric Assessment in Hospitalized Older Patients. *Rejuvenation Research*. 2008 Feb;11(1):151–61.

[4] Cederholm T, Jensen GL, Correia MITD, Gonzalez MC, Fukushima R, Higashiguchi T, et al. GLIM criteria for the diagnosis of malnutrition - A consensus report from the global clinical nutrition community. *Clinical nutrition* (Edinburgh, Scotland) [Internet]. 2019;38(1):1–9. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/30181091>

[5] Dialysis Unit, Ospedale Sant'Anna di Como.

Elderly hemodialysis patient dataset.

Unpublished internal dataset, provided for academic evaluation.

Appendix A

Appendix A.1 Regression Diagnostics

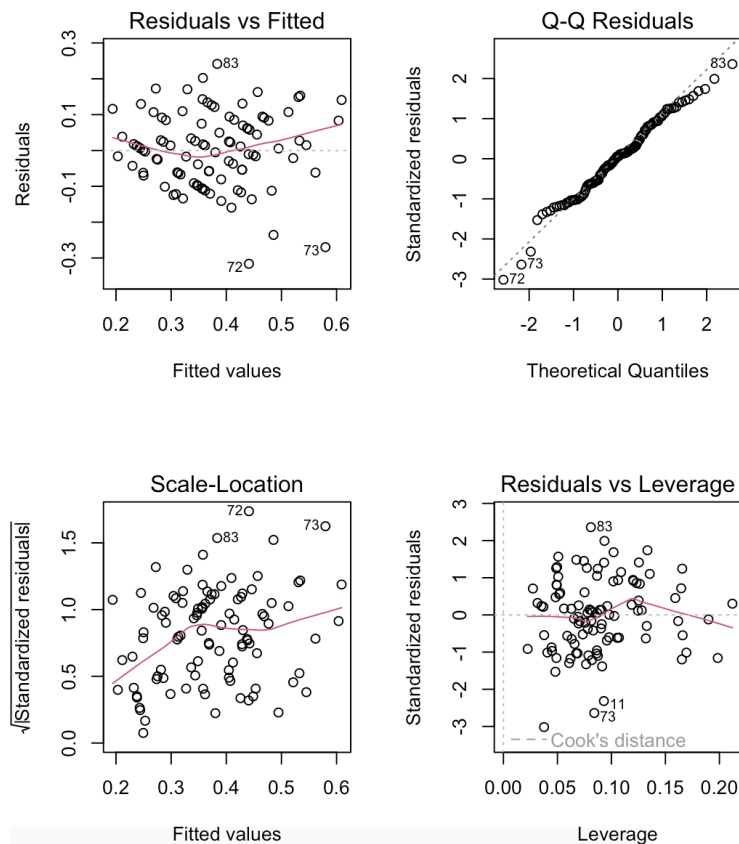


Figure A1. Residuals versus fitted values.

Figure A2. Normal Q-Q plot of residuals.

Figure A3. Scale–location plot.

Figure A4. Residuals versus leverage.

Appendix A.2 Additional Statistical Output

Output A1. Summary of multiple linear regression model

```
Call:
lm(formula = `Punteggio MPI` ~ `GLIM EXPAND FINAL` + `LTI Kg/mt2` +
  `Fat (%)` + Albumina + `Kt/V` + `eta 2023` + session + CIRS,
  data = df_cc)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.316062	-0.067006	0.006252	0.083291	0.241506

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.919448	0.315254	2.917	0.004447 **
`GLIM EXPAND FINAL`	0.013841	0.016126	0.858	0.392952
`LTI Kg/mt <sup>2</sup> `	-0.031935	0.008023	-3.981	0.000137 ***
`Fat (%)`	-0.004429	0.001862	-2.378	0.019470 *
Albumina	-0.083157	0.036518	-2.277	0.025093 *
`Kt/V`	0.001098	0.047709	0.023	0.981688
`eta 2023`	0.003604	0.001898	1.899	0.060693 .
session	-0.020759	0.030895	-0.672	0.503319
CIRS	0.005249	0.003503	1.498	0.137508

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1068 on 92 degrees of freedom

Multiple R-squared: 0.4566, Adjusted R-squared: 0.4094

F-statistic: 9.664 on 8 and 92 DF, p-value: 1.235e-09

Appendix A.3 R Code

note : possible sub-question: